3. Bayesian Networks: Inference

CSC3043S Reasoning in AI: Bayesian Networks

Dr Jan Buys

University of Cape Town

[Based on slides created by Dan Klein and Pieter Abbeel for CS188 Intro to AI at UC Berkeley. All materials available at http://ai.berkeley.edu.]

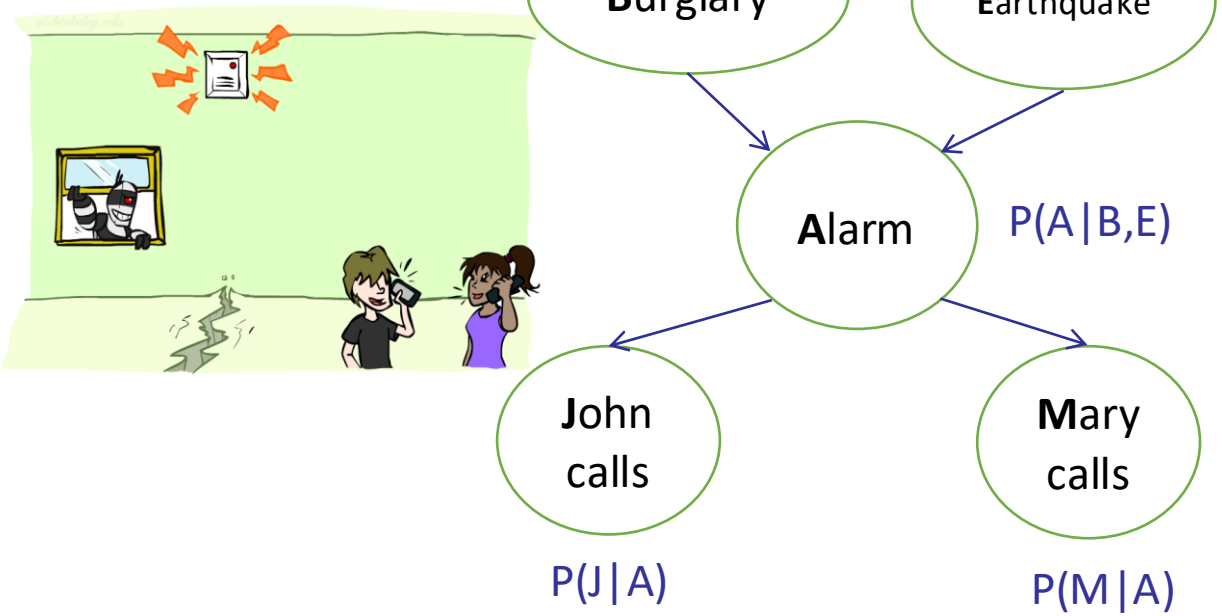


UNIVERSITY OF CAPE TOWN

IYUNIVESITHI YASEKAPA • UNIVERSITEIT VAN KAAPSTAD

Inference

- Example: Alarm network



What is:

- P(M)
- P(B|E)
- P(B|M)

Inference

Inference by enumeration:

1. **Select** the probability table entries consistent with the evidence:

$$P(Q, H_1, H_2, \dots, H_r, e_1, e_2, \dots, e_k)$$

2. **Marginalize** the hidden variables:

$$P(Q, e_1, e_2, \dots, e_k) = \sum_{h_1, \dots, h_r} P(Q, h_1, \dots, h_r, e_1, e_2, \dots, e_k)$$

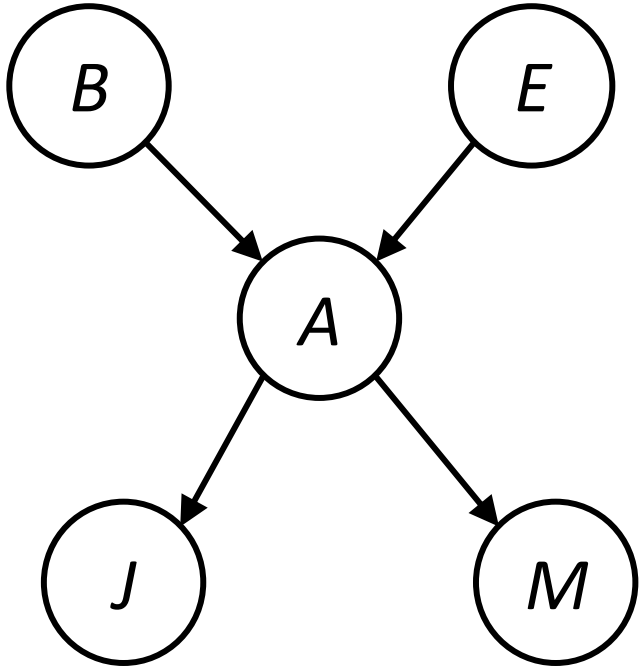
3. **Normalize** the distribution:

$$Z = \sum_{q} P(q, e_1, e_2, \dots, e_k)$$

$$P(Q|e_1, e_2, \dots, e_k) = \frac{1}{Z} P(Q, e_1, e_2, \dots, e_k)$$

Inference

- Inference by enumeration in a Bayesian network:



$$P(B \mid +j, +m) \propto_B P(B, +j, +m)$$ **1. Select**

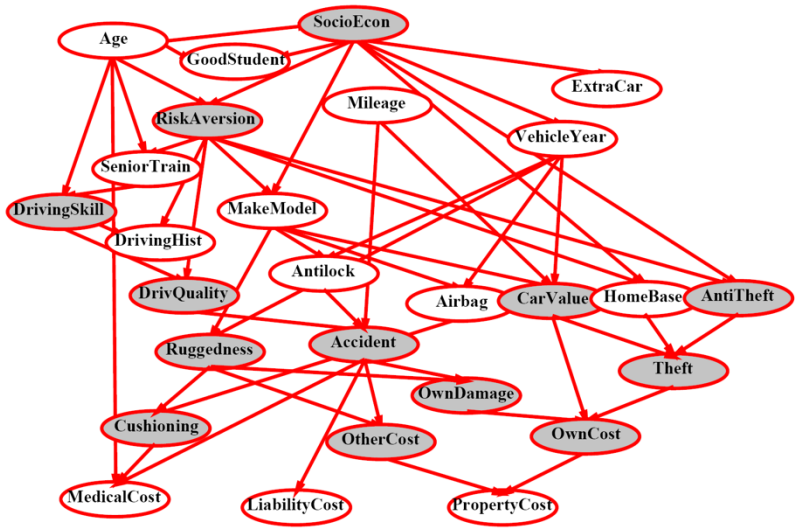
$$= \sum_{e,a} P(B, e, a, +j, +m)$$ **2. Marginalize**

$$= \sum_{e,a} P(B)P(e)P(a|B,e)P(+j|a)P(+m|a)$$ **3. Normalize** $\times \frac{1}{Z}$

$$= P(B)P(+e)P(+a|B,+e)P(+j|+a)P(+m|+a) + P(B)P(+e)P(-a|B,+e)P(+j|-a)P(+m|-a)$$
$$P(B)P(-e)P(+a|B,-e)P(+j|+a)P(+m|+a) + P(B)P(-e)P(-a|B,-e)P(+j|-a)P(+m|-a)$$

Inference

- In a network with a large number of variables, inference by enumeration is not tractable
- e.g. $P(Antilock|observed\ variables) = ?$
- The whole joint distribution is computed (joined up) before summing out (marginalizing) the hidden variables
- Instead, it is more efficient to interleave joining and marginalizing



Inference

Reframe inference by enumeration using *factors*:

- A **factor** is any quantity based on a probability distribution
- Initial factors are local CPTs

$P(R)$

+r	0.1
-r	0.9

$P(T|R)$

+r	+t	0.8
+r	-t	0.2
-r	+t	0.1
-r	-t	0.9

$P(L|T)$

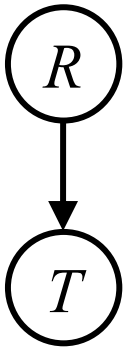
+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9

Inference

Operation 1: Join factors

- Just like a database join
- Get all factors over joining variable
- Build a new factor over the union of the variable involved

- Example: Join on R
  - Each entry is a pointwise product: $\forall r, t: \quad P(r,t) = P(r) \cdot P(t|r)$



$P(R) \times P(T|R)$ → $P(R,T)$

+r	0.1
-r	0.9

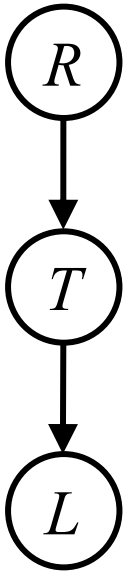
+r	+t	0.8
+r	-t	0.2
-r	+t	0.1
-r	-t	0.9

+r	+t	0.08
+r	-t	0.02
-r	+t	0.09
-r	-t	0.81



Inference

- Example: Multiple joints



$P(R)$

+r	0.1
-r	0.9

$P(T|R)$

+r	+t	0.8
+r	-t	0.2
-r	+t	0.1
-r	-t	0.9

$P(L|T)$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9

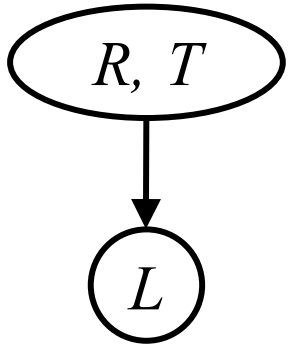
Join R

$P(R,T)$

+r	+t	0.08
+r	-t	0.02
-r	+t	0.09
-r	-t	0.81

$P(L|T)$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9



Join T



$P(R,T,L)$

+r	+t	+l	0.024
+r	+t	-l	0.056
+r	-t	+l	0.002
+r	-t	-l	0.018
-r	+t	+l	0.027
-r	+t	-l	0.063
-r	-t	+l	0.081
-r	-t	-l	0.729

Inference

Operation 2: Eliminate

- Take a factor and sum out (marginalize) a variable
- Shrinks a factor to a smaller one
- A projection operation

- Example:

$P(R,T)$

+r	+t	0.08
+r	-t	0.02
-r	+t	0.09
-r	-t	0.81

sum R 

$P(T)$

+t	0.17
-t	0.83

Inference

- Example: Multiple elimination

$P(R, T, L)$

R, T, L

+r	+t	+l	0.024
+r	+t	-l	0.056
+r	-t	+l	0.002
+r	-t	-l	0.018
-r	+t	+l	0.027
-r	+t	-l	0.063
-r	-t	+l	0.081
-r	-t	-l	0.729

Sum out R

T, L

+t	+l	0.051
+t	-l	0.119
-t	+l	0.083
-t	-l	0.747

Sum out T

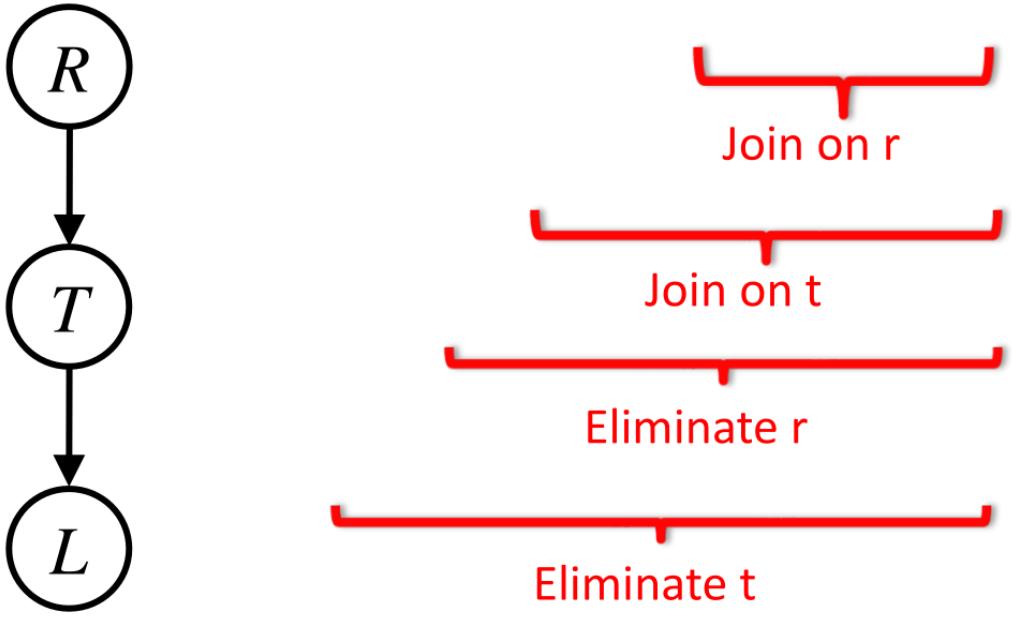
L

+l	0.134
-l	0.886

Inference

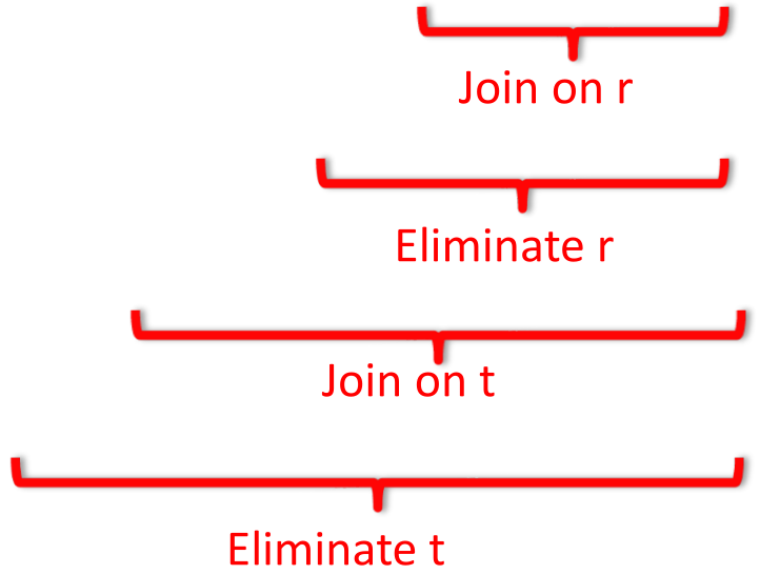
- **Inference by enumeration**: Multiple joins, then multiple eliminates

$$P(L) = \sum_t \sum_r P(L|t)P(r)P(t|r)$$

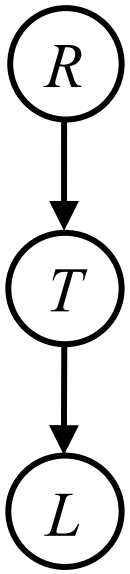


- **Variable elimination**: Interleave join and eliminate

$$P(L) = \sum_t P(L|t) \sum_r P(r)P(t|r)$$



Variable Elimination example



$P(R)$

+r	0.1
-r	0.9

$P(T|R)$

+r	+t	0.8
+r	-t	0.2
-r	+t	0.1
-r	-t	0.9

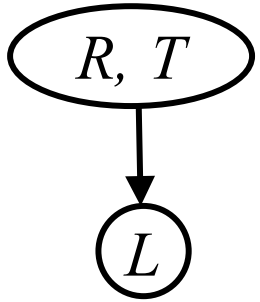
$P(L|T)$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9

Join R

$P(R,T)$

+r	+t	0.08
+r	-t	0.02
-r	+t	0.09
-r	-t	0.81



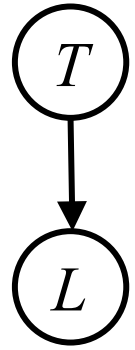
$P(L|T)$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9

Sum out R

$P(T)$

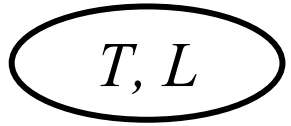
+t	0.17
-t	0.83



$P(L|T)$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9

Join T



$P(T,L)$

+t	+l	0.051
+t	-l	0.119
-t	+l	0.083
-t	-l	0.747

Sum out T



$P(L)$

+l	0.134
-l	0.866

Variable Elimination

Handling evidence:

- If there is evidence, first get factors that select the evidence
- Example: With no evidence the initial factors are:

$P(R)$

+r	0.1
-r	0.9

$P(T|R)$

+r	+t	0.8
+r	-t	0.2
-r	+t	0.1
-r	-t	0.9

$P(L|T)$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9

- For computing $P(L|+r)$ the initial factors become:

$P(+r)$

+r	0.1

$P(T|+r)$

+r	+t	0.8
+r	-t	0.2

$P(L|T)$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9

Variable Elimination

Handling evidence (continued):

- Eliminate all hidden variables (variables other than query and evidence)
- Result will be a selected joint of query and evidence
- To get answer, just normalize this

- Example for $P(L|+r)$:

$P(+r, L)$

+r	+l	0.026
+r	-l	0.074

Normalize



$P(L|+r)$

+l	0.26
-l	0.74

Variable Elimination

General procedure for variable elimination:

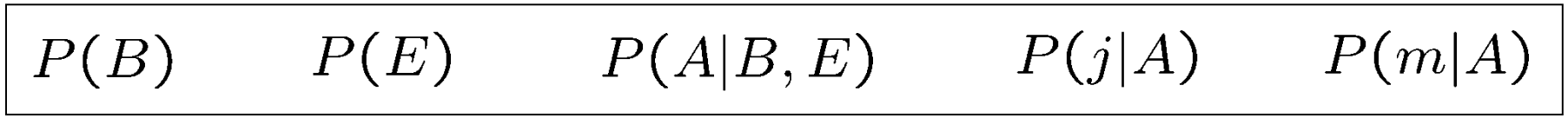
- Query $P(Q|E_1 = e_1, \ldots E_k = e_k)$
- Start with initial factors: Local CPTs instantiated with evidence
- While there are still hidden variables (not Q or evidence):
  - Pick a hidden variable H
  - Join all factors mentioning H
  - Eliminate (marginalize) H
- Join all remaining factors and normalize

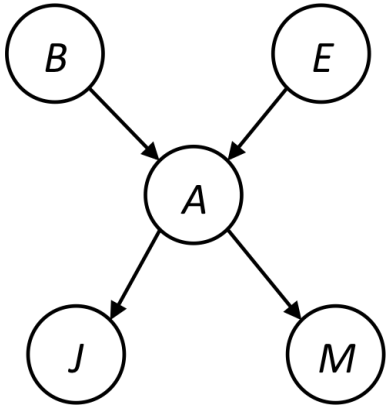
Variable Elimination

- Alarm network example:

$$P(B|j,m) \propto P(B,j,m)$$

- Initial factors:





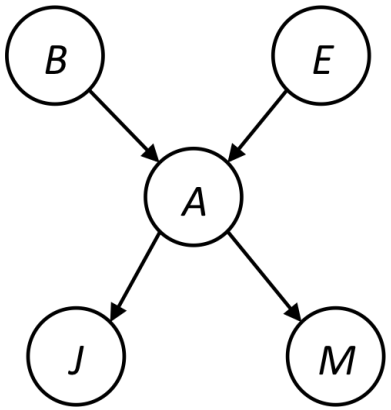
- Choose variable A:

$P(A|B,E)$
$P(j|A)$
$P(m|A)$

Join ⨯ $P(j,m,A|B,E)$ Eliminate $\sum$ $P(j,m|B,E)$

Variable Elimination

Remaining factors: $P(B)$ $P(E)$ $P(j,m|B,E)$



- Choose E:

$P(E)$, $P(j,m|B,E)$ → × → $P(j,m,E|B)$ → $\sum$ → $P(j,m|B)$

$P(B)$ $P(j,m|B)$

- Finally choose variable B:

$P(B)$, $P(j,m|B)$ → × → $P(j,m,B)$ → Normalize → $P(B|j,m)$

Variable Elimination

- Same example in equations:

$$P(B|j,m) \propto P(B,j,m)$$

$P(B)$	$P(E)$	$P(A\|B,E)$	$P(j\|A)$	$P(m\|A)$

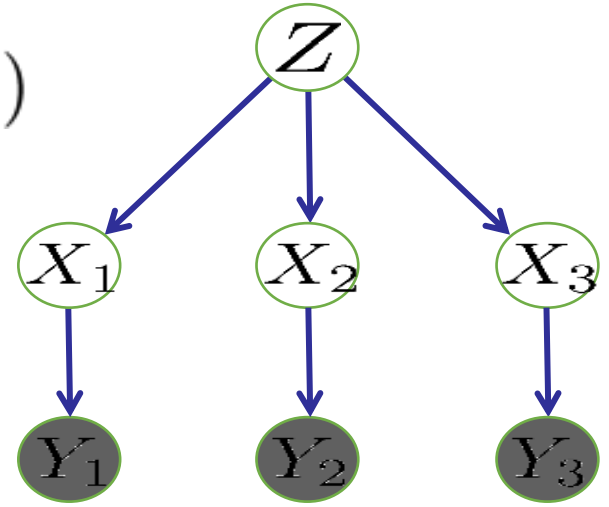
$$
\begin{aligned}
P(B|j,m) \propto\ & P(B,j,m) && \text{Obtain marginal from joint by summing out} \\
=\ & \sum_{e,a} P(B,j,m,e,a) \\
=\ & \sum_{e,a} P(B)P(e)P(a|B,e)P(j|a)P(m|a) && \text{use Bayes' net joint distribution expression} \\
=\ & \sum_{e} P(B)P(e)\sum_{a} P(a|B,e)P(j|a)P(m|a) && \text{use x*(y+z) = xy + xz} \\
=\ & \sum_{e} P(B)P(e)f_1(B,e,j,m) && \text{joining on a, and then summing out gives } f_1 \\
=\ & P(B)\sum_{e} P(e)f_1(B,e,j,m) && \text{use x*(y+z) = xy + xz} \\
=\ & P(B)f_2(B,j,m) && \text{joining on e, and then summing out gives } f_2
\end{aligned}
$$

All we are doing is exploiting uwy + uwz + uxy + uxz + vwy + vwz + vxy +vxz = (u+v)(w+x)(y+z) to improve computational efficiency

Variable Elimination

- The order in which variables are eliminated will affect the computational complexity
- Computational complexity critically depends on the size of the largest factor (number of table entries) being generated

Example query: $P(X_3|Y_1 = y_1, Y_2 = y_2, Y_3 = y_3)$

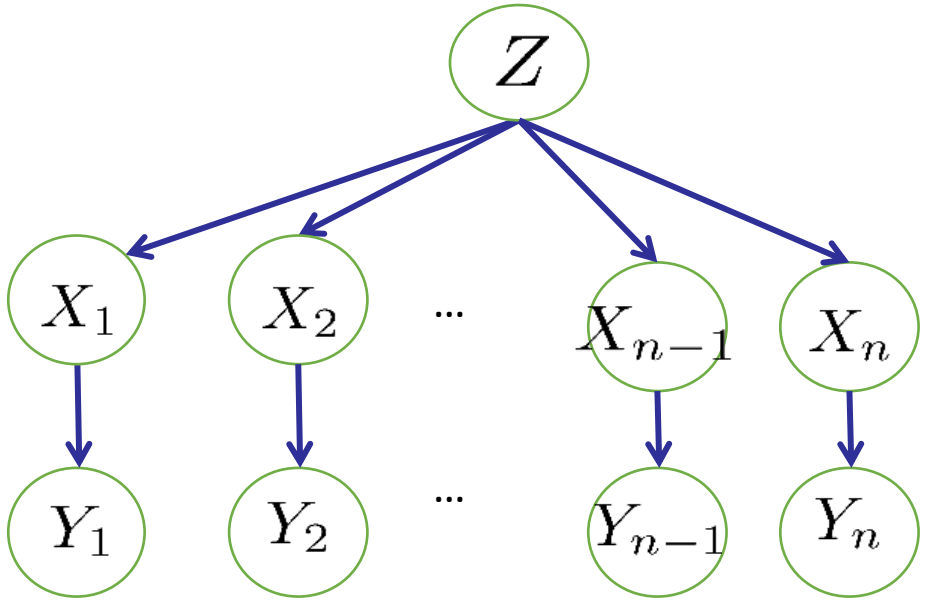


- For elimination order X_1, X_2, Z all generated factors have size two as they all only have one variable
- If we eliminate Z first, have a factor of size 4

Variable Elimination

For query $P(X_n | y_1,\ldots,y_n)$:

- Elimination order $Z, X_1, \ldots, X_{n-1}$
  - n+1 variables in table (2^{n+1}entries)
- Elimination order $X_1, \ldots, X_{n-1}, Z$
  - 2 variables in each table (2^2 entries)



- In general, we can't guarantee that there will exist an ordering that only results in small factors